Milestone 2 (Technical Artifact) — Research Project

Performance Analysis of ISAC-PASS under EDMA Geometry

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Abstract

Pinching-antenna systems (PASS) let the radiation point of an antenna move along a dielectric waveguide, which makes them attractive for integrated sensing and communication (ISAC). Blockage-aware PASS studies consider communication only, and blockage-aware ISAC studies consider conventional antennas only. The technical artifact of this milestone is an analytical framework and a matching simulator for a multi-segment ISAC-PASS under environment-division multiple access (EDMA) geometry, where line-of-sight (LoS) blockage of the cross-segment links governs interference, clutter, and leakage. The artifact provides (i) a closed-form mean cross-segment LoS probability $p_1(\phi)$, (ii) closed-form lower bounds on the ergodic sum communication rate (CR) and sensing rate (SR) for any number of segments, (iii) a high-SNR condition under which EDMA outperforms TDMA, and (iv) a Monte Carlo (MC) simulator, verified against the closed forms. The full performance evaluation is planned for M3.

1 Introduction and Milestone Scope

ISAC lets one network share spectrum and hardware between data transmission and radar-like sensing [1, 2]. In PASS, a small dielectric "pinch" placed on a waveguide makes it radiate almost anywhere along it, so the antenna can be brought close to a user or target [3,4]. In indoor and industrial settings, however, LoS links are often blocked. This project asks how such blockage affects a PASS that serves users and senses targets at the same time.

The project has four milestones: M1 (proposal and scoping), **M2 (technical artifact)**, M3 (results and evaluation), and M4 (final submission). This report covers M2. The artifact consists of the system model, the analytical derivations, and the simulation code in `Code/src`. Only the checks needed to show that the artifact is correct are reported here; the complete evaluation is left to M3.

2 Literature Synthesis and Technical Gap

Prior work can be placed along three axes: the antenna architecture, the function (communication only or ISAC), and the propagation assumption (always LoS or blockage-aware), as summarised in Table 1.

PASS for ISAC. Ouyang *et al.* [5] characterise the CR–SR rate region of PASS using the mutual-information sensing framework of [6]. Jiang *et al.* [7] split the waveguide into isolated, separately fed segments (SWAN), so that several user–target pairs can be served. Both assume that every link is LoS, so cross-segment interference and clutter are always present. **Blockage.** Ding and Poor [8] model a link of length r as LoS with probability $e^{-\phi r^2}$, and show that for PASS, blockage can *help*, because long interfering links are blocked more often than short desired ones. EDMA [9] turns this into a multiple-access scheme with each antenna at its user, but for communication only. Sun *et al.* [10] analyse blockage in ISAC networks, but with conventional antennas.

Table 1: Positioning of prior work along the axes of the problem.

	Always LoS	**Blockage-aware, comm. only**	**Blockage-aware, ISAC**
Conventional antennas	classical ISAC [1]	—	Sun *et al.* [10]
PASS, single waveguide	Ouyang *et al.* [5]	Ding & Poor [8], EDMA [9]	—
PASS, segmented	SWAN [7]	—	**This project**

Gap. *No existing work analyses how LoS blockage of the cross-segment links, which carry communication interference, sensing clutter, and sensing leakage at the same time, affects the communication*

and sensing rates of a multi-segment ISAC-PASS, or when concurrent EDMA operation is better than TDMA. Combining existing results does not close this gap, for three reasons. First, sensing adds clutter and leakage, which follow different distance laws. Second, an antenna that serves both a user and a target sits at a random point that depends on both, which changes the cross-link distance distribution. Third, both SINRs are ratios of random terms, so exact ergodic rates are intractable.

3 Problem Formulation and System Model

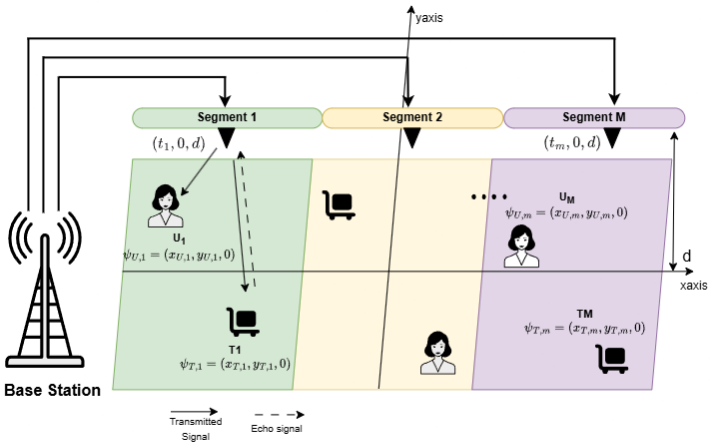


Figure 1: Network model: M segments, each with one pinching antenna at the midpoint of its user and target.

Geometry. A base station feeds a lossless waveguide of length D_L at height d, split as in [7] into M isolated segments of width $S = D_L/M$, each with its own RF chain (Fig. 1). Segment m serves a user U_m and a target T_m, both uniformly distributed within it, through one pinching antenna PA_m. The antenna is placed at the midpoint $t_m = (x_{U,m} + x_{T,m})/2$. A link of length r is unblocked ($\delta = 1$) with probability $\Pr\{\delta = 1\} = e^{-\phi r^2}$ [8]. Interference reaches segment m only through unblocked cross links from $\mathcal{N}_m = \{1, \ldots, M\} \setminus \{m\}$.

Communication. With free-space channels $h_{i,m} = \sqrt{\eta}\, e^{-j2\pi r_{i,m}/\lambda}/r_{i,m}$, $\eta = c^2/(16\pi^2 f_c^2)$, and all segments transmitting simultaneously with power P, the SINR and rate of U_m are

$$\gamma_m = \frac{\rho |h_{m,m}|^2}{1 + \rho \sum_{i \in \mathcal{N}_m} \delta_{i,m} |h_{i,m}|^2}, \qquad R_m = \log_2(1 + \gamma_m), \qquad \rho = P/\sigma^2. \tag{1}$$

Sensing. PA_m senses T_m monostatically with a unit-modulus probe of length L. The target reflection follows the Swerling-I model $\beta_i \sim \mathcal{CN}(0, \alpha_s)$ [5, 11]. Besides the desired echo, PA_m receives *clutter* (its own probe reflected by T_i) and *leakage* (PA_i's probe reflected by T_i). Both arrive over the link PA_m–T_i and therefore share its blockage state $\delta^{\mathrm{s}}_{m,i}$. After a matched filter, treating clutter and leakage as Gaussian noise [6], the sensing SINR and rate are

$$\Gamma_m = \frac{\rho_s \mu_{m,m}}{1 + \rho_s \sum_{i \in \mathcal{N}_m} \delta^{\mathrm{s}}_{m,i} \left(\mu_{m,i} + \nu_{m,i}/L\right)}, \qquad R_{s,m} = \tfrac{1}{L} \log_2(1 + \Gamma_m), \tag{2}$$

where $\rho_s = PL/\sigma_s^2$, and the clutter and leakage gains are

$$\mu_{m,i} = \frac{\alpha_s \eta^2}{(r^{\mathrm{s}}_{m,i})^4}, \qquad \nu_{m,i} = \frac{\alpha_s \eta^2}{(r^{\mathrm{s}}_{i,i})^2 (r^{\mathrm{s}}_{m,i})^2}.$$

Research questions. Q1: What is the probability $p_1(\phi)$ that a cross-segment link is LoS? **Q2:** Can the ergodic sum rates $\bar{R}_c = \mathbb{E}[\sum_m R_m]$ and $\bar{R}_s = \mathbb{E}[\sum_m R_{s,m}]$ be lower bounded in closed form for any M? **Q3:** When does EDMA outperform TDMA, and how should M be chosen?

4 Methodology: The Technical Artifact

4.1 Analytical Framework

For tractability, nodes are placed under the waveguide ($y = 0$), the short own links are LoS, and impairments come only from the adjacent segments $\mathcal{N}_m^{\mathrm{nb}} = \{m-1, m+1\}$. All three simplifications are checked numerically.

Step 1: mixture bound. Let $\mathcal{A}_m$ be the event that at least one adjacent cross link is unblocked. If all are blocked, the rate is the interference-free rate $R_{\mathrm{cl}} = \log_2\big(1 + \rho\eta/(w^2+d^2)\big)$, where w is the PA–user offset. Hence $R_m \geq R_{\mathrm{cl}} - R_{\mathrm{cl}}\mathbf{1}\{\mathcal{A}_m\}$. Since w is triangular on $[-S/2, S/2]$, the first term is averaged exactly with

$$I(a) \triangleq \mathbb{E}_w[\log_2(w^2+a)] = \tfrac{1}{\ln 2}\Big[\big(1-\tfrac{4a}{S^2}\big)\ln\big(\tfrac{S^2}{4}+a\big) + \tfrac{4a}{S^2}\ln a - 3 + \tfrac{8\sqrt{a}}{S}\tan^{-1}\big(\tfrac{S}{2\sqrt{a}}\big)\Big], \tag{3}$$

which gives $B_{\mathrm{c}} = I(d^2+\rho\eta) - I(d^2)$. For sensing, $(w^2+d^2)^2+\Omega^2$ factors into complex-conjugate terms, so the same $I(a)$ evaluated at complex a gives $B_{\mathrm{s}} = \frac{2}{L}[\mathrm{Re}\{I(d^2+j\Omega)\} - I(d^2)]$, with $\Omega = \eta\sqrt{\rho_s\alpha_s}$.

Step 2: cross-link LoS probability (Q1). The normalised distance between a user and the neighbouring antenna has the piecewise-quadratic density $f_V(v) = 2v^2$, $1-2(v-1)^2$, and $2(2-v)^2$ on $[0, \frac{1}{2}]$, $[\frac{1}{2}, \frac{3}{2}]$, and $[\frac{3}{2}, 2]$, respectively. Averaging the blockage model over V gives

$$p_1(\phi) = e^{-\phi d^2}\int_0^2 f_V(v)\, e^{-\phi S^2 v^2}\, dv, \tag{4}$$

which reduces to closed-form Gaussian integrals involving $\mathrm{erf}(\cdot)$. For wide segments,

$$p_1(\phi) \approx \frac{\sqrt{\pi}\, e^{-\phi d^2}}{2\phi^{3/2}S^3},$$

i.e., the cross-link LoS probability decays with the cube of the segment width.

Step 3: rate bounds (Q2). Bounding $R_{\mathrm{cl}} \leq B_{\mathrm{c}}^{\max} = \log_2(1+\rho\eta/d^2)$ and applying the union bound $\Pr\{\mathcal{A}_m\} \leq |\mathcal{N}_m^{\mathrm{nb}}|\, p_1$, which holds for any correlation between blockage states, gives

Theorem 1. $\bar{R}_c \geq \sum_{m=1}^{M}\Big[B_{\mathrm{c}} - B_{\mathrm{c}}^{\max}|\mathcal{N}_m^{\mathrm{nb}}|\, p_1(\phi)\Big]^+$, *where* $[x]^+ = \max\{x, 0\}$.

Theorem 2. *The same bound holds for* $\bar{R}_s$, *with* B_{s} *and* $B_{\mathrm{s}}^{\max} = \frac{1}{L}\log_2(1+\rho_s\alpha_s\eta^2/d^4)$ *in place of* B_{c} *and* $B_{\mathrm{c}}^{\max}$.

Step 4: EDMA vs. TDMA (Q3). TDMA has no interference, so its sum rate grows as $\log_2 P$. EDMA gains a $\log_2 P$ slope only from segments whose cross links are all blocked, so its slope is at least $M - 2(M-1)p_1$. EDMA therefore outperforms TDMA at high SNR if $p_1(\phi) < \frac{1}{2}$ (also necessary for $M = 2$), which gives a design rule for choosing M.

4.2 Simulation Artifact

The simulator `edma_sim.py` (Python/NumPy) draws 10^5 random user/target placements. For each placement it evaluates the blockage expectation *exactly*, by enumerating every blockage pattern of the cross links, so there is no blockage sampling noise. It implements EDMA with the antenna at the midpoint, at the user (as in [9]), and at the target, as well as a TDMA benchmark: one RF chain with power MP, antennas at the segment centres, and bistatic sensing through a second receive waveguide. It also implements every closed form ($I(a)$, p_1, and both bounds). Results are cached under a hash of all model parameters, so stale data cannot be reused. `plot_all.py` generates the figures, and `verify_lb.py` independently re-derives each closed form. Parameters: $f_c = 28$ GHz, $d = 3$ m, $D_L = 40$ m, $\sigma^2 = \sigma_s^2 = -90$ dBm [8], $L = 5$, $\alpha_s = 10$ [5], $\phi = 0.02$ [9], $P \in [0, 30]$ dBm.

Table 2: Closed forms vs. independent numerical evaluation (`verify_lb.py`, $P = 30$ dBm, $D_L = 40$ m).

	$M = 2$ ($S = 20$ m)		$M = 5$ ($S = 8$ m)	
Quantity	Closed form	Numerical	Closed form	Numerical
B_c (quadrature)	15.1405	15.1405	15.9714	15.9714
$p_1(\phi)$ (brute-force MC, 2×10^6 samples)	0.0320	0.0319	0.2779	0.2779
Sum CR: Theorem 1 vs. MC simulation	29.24	29.45	43.61	51.43

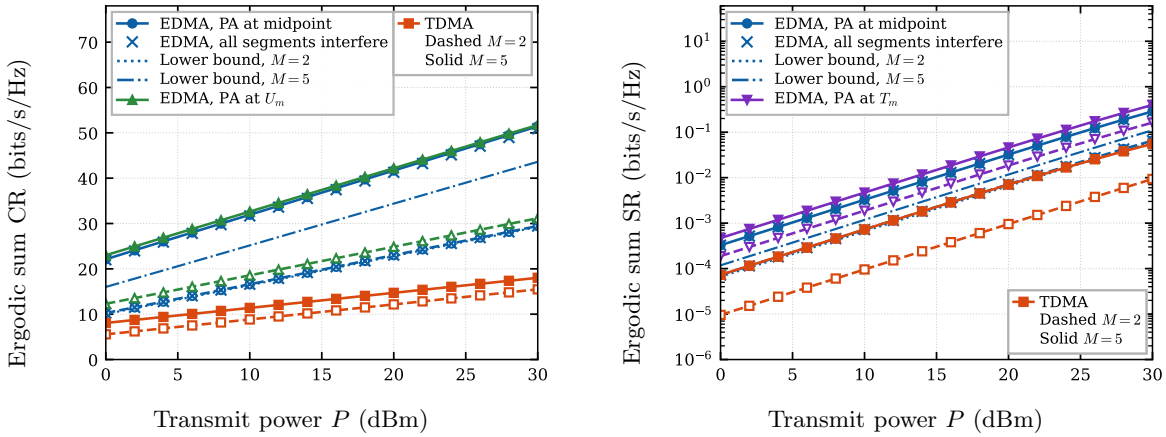


Figure 2: Simulator output vs. the bounds of Theorems 1–2 ($D_L = 40$ m; dashed/hollow: $M = 2$, solid: $M = 5$).

5 Artifact Verification

Table 2 shows that every closed-form ingredient matches an independent numerical evaluation, and that the CR bound lies below the simulated rate, as Theorem 1 requires. It is tight for $M = 2$ and looser for $M = 5$, where the union bound covers more neighbours. Fig. 2 confirms that both bounds stay below the simulator at every power. Including all segments instead of only adjacent ones (crosses) changes the rates by at most 1%, which supports the adjacent-only simplification. As a sanity check of the TDMA analysis, the simulated EDMA/TDMA CR ratio for $M = 2$ and 5 (1.85–1.90 and 2.74–2.85) matches the predicted gains $M - 2(M-1)p_1$ of 1.94 and 2.78. The artifact is therefore ready for the systematic evaluation in M3.

6 SOTA Positioning

Table 3: Capability comparison with the closest works.

Work	PASS	ISAC	Blockage	Concurrent segments	Clutter & leakage	Closed-form rates
Ding & Poor [8]	✓	×	✓	×	×	✓
EDMA [9]	✓	×	✓	✓	×	✓
Ouyang *et al.* [5]	✓	✓	×	×	×	—
SWAN [7]	✓	✓	×	✓	—	—
Sun *et al.* [10]	×	✓	✓	×	—	✓
This project	✓	✓	✓	✓	✓	✓

Existing works give closed-form PASS rates under blockage for communication only [8,9], or PASS-ISAC characterisations that assume every link is LoS [5,7]. This project differs in three ways (Table 3): (i) a closed-form cross-link LoS probability for the midpoint geometry, (ii) clutter and leakage modelled separately inside the SR bound, and (iii) a design rule, $p_1(\phi) < \frac{1}{2}$ together with the S^{-3} scaling, for choosing M.

7 Plan for M3 and M4

M3 (results and evaluation): rates versus P and M for several D_L; comparison of placing the antenna at the midpoint, at the user, and at the target; validation of the $p_1 < \frac{1}{2}$ design rule. **M4 (final submission):** a physically grounded target RCS, a conventional-antenna benchmark, and the final write-up.

Individual contribution. All work in this single-member project, from the literature review and modelling to the derivations, code, and report, was carried out by Khushi Shah under the mentorship of Prof. Dhaval Patel.

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